BASIC FORMULAS

Schrödinger Equation

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi \qquad \hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}$$

Stationary States

$$\hat{H}\psi = E\psi \qquad \Psi = \psi e^{-iEt/\hbar}$$

Operators

$$\hat{\boldsymbol{r}} = \boldsymbol{r} \qquad \hat{\boldsymbol{p}} = -i\hbar\boldsymbol{\nabla}$$

$$\left[\hat{r}_i, \hat{p}_j\right] = i\hbar\,\delta_{ij}$$

Uncertainty Principle

$$\sigma_A\sigma_B \geq \left|\frac{1}{2i}\langle[A,B]\rangle\right|$$

$$\sigma_x\sigma_p \geq \frac{\hbar}{2} \qquad \Delta t\sigma_H \geq \frac{\hbar}{2}$$

STATISTICS

$$\left\langle f\middle|\hat{Q}\middle|g\right\rangle = \int_{-\infty}^{\infty} dx\, f^*(x)\,\hat{Q}\,g(x) = \underline{f}^{\dagger}\cdot\underline{Q}\cdot\underline{g}$$

$$\langle Q(x,p)\rangle = \langle\Psi|Q(\hat{x},\hat{p})|\Psi\rangle \qquad \sigma_f = \sqrt{\langle f^2\rangle - \langle f\rangle^2}$$

INTEGRALS

$$\int_a^b f\frac{dg}{dx}dx = fg\Big|_a^b - \int_a^b \frac{df}{dx}g\,dx \qquad \int_0^\infty x^n e^{-x/a}\,dx = n!\,a^{n+1}$$

$$\int_0^\infty x^{2n}e^{-x^2/a^2}\,dx = \sqrt{\pi}\frac{(2n)!}{n!}\left(\frac{a}{2}\right)^{2n+1} \qquad \int_0^\infty x^{2n+1}e^{-x^2/a^2}\,dx = \frac{n!}{2}a^{2n+2}$$

REPRESENTATIONS

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dp\, e^{ipx/\hbar}\,\Phi(p,t)$$

$$\Phi(p,t) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx\, e^{-ipx/\hbar}\,\Psi(x,t)$$

$$\Psi(x,t) = \sum_n c_n(t)\psi_n(x)$$

$$c_n(t) = c_n e^{-iE_nt/\hbar} = \langle n|\Psi(t)\rangle = \int_{-\infty}^{\infty} dx\,\psi_n^*(x)\,\Psi(x,t)$$

$$\langle n|m\rangle = \delta_{nm} \qquad \sum_n |n\rangle\langle n| = I$$

$$\langle x|x'\rangle = \delta(x-x') \qquad \int dx\,|x\rangle\langle x| = I$$

ANGULAR MOMENTUM

$$[J_x,J_y] = i\hbar J_z \qquad [J_y,J_z] = i\hbar J_x \qquad [J_z,J_x] = i\hbar J_y$$

$$J_\pm = J_x \pm iJ_y$$

$$[J^2,\boldsymbol{J}] = 0 \qquad [J_z,J_\pm] = \pm\hbar J_\pm \qquad [J_+,J_-] = 2\hbar J_z$$

$$J^2|j\;m\rangle = \hbar^2 j(j+1)|j\;m\rangle \qquad J_z|j\;m\rangle = m\hbar|j\;m\rangle$$

$$J_+|j\;m-1\rangle = d_m|j\;m\rangle \qquad J_-|j\;m\rangle = d_m|j\;m-1\rangle$$

$$d_m = \hbar\sqrt{(j-m+1)(j+m)}$$

Spin 1/2

$$\underline{\sigma_x} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \qquad \underline{\sigma_y} = \begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} \qquad \underline{\sigma_z} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \qquad \underline{\boldsymbol{S}} = \frac{\hbar}{2}\underline{\boldsymbol{\sigma}}$$

$$\underline{S^2} = \frac{3}{4}\hbar^2\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} \qquad \underline{S_+} = \hbar\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} \qquad \underline{S_-} = \hbar\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}$$

COMMON PROBLEMS

Free Particle

$$\psi(x) = Ae^{\pm ikx} \qquad E = \frac{\hbar^2 k^2}{2m}$$

Infinite Square Well

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \qquad E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

Harmonic Oscillator

$$\psi_n(\xi) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} H_n(\xi) e^{-\frac{1}{2}\xi^2} \qquad E_n = \left(n + \frac{1}{2}\right)\hbar\omega \qquad \xi = \sqrt{\frac{m\omega}{\hbar}}x$$

$$a = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a) \qquad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(a^\dagger - a)$$

$$[a, a^\dagger] = 1$$

$$[\hat{H}, a^\dagger] = \hbar\omega a^\dagger \qquad [\hat{H}, a] = -\hbar\omega a$$

$$\hat{H} = \left(a^\dagger a + \frac{1}{2}\right)\hbar\omega$$

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \qquad a|n\rangle = \sqrt{n}|n-1\rangle$$

Hydrogen Atom

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi) \qquad E_n = -\frac{mc^2\alpha^2}{2}\frac{1}{n^2} \approx \frac{-13.6 \text{ eV}}{n^2}$$

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} = \frac{\hbar}{\alpha mc} \qquad \alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$$

SPECIAL FUNCTIONS

Hermite Polynomials

$$H_0(\xi) = 1 \qquad H_3(\xi) = 8\xi^3 - 12\xi$$

$$H_1(\xi) = 2\xi \qquad H_4(\xi) = 16\xi^4 - 48\xi^2 + 12$$

$$H_2(\xi) = 4\xi^2 - 2 \qquad H_5(\xi) = 32\xi^5 - 160\xi^3 + 120\xi$$

Spherical Harmonics

$$Y_0^0 = \sqrt{\frac{1}{4\pi}} \qquad Y_2^0 = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta \qquad Y_2^{\pm 1} = \mp\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta\, e^{\pm i\phi}$$

$$Y_1^{\pm 1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\phi} \qquad Y_2^{\pm 2} = \sqrt{\frac{15}{32\pi}}\sin^2\theta\, e^{\pm 2i\phi}$$

Radial Wave Functions

$$R_{10} = 2a_0^{-3/2} e^{-r/a_0} \qquad R_{30} = \frac{2}{\sqrt{27}} a_0^{-3/2}\left(1 - \frac{2r}{3a_0} + \frac{2}{27}\left(\frac{r}{a_0}\right)^2\right) e^{-r/3a_0}$$

$$R_{20} = \frac{1}{\sqrt{2}} a_0^{-3/2}\left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0} \qquad R_{31} = \frac{8}{27\sqrt{6}} a_0^{-3/2}\left(1 - \frac{r}{6a_0}\right)\left(\frac{r}{a_0}\right) e^{-r/3a_0}$$

$$R_{21} = \frac{1}{\sqrt{24}} a_0^{-3/2} \frac{r}{a_0} e^{-r/2a_0} \qquad R_{32} = \frac{4}{81\sqrt{30}} a_0^{-3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/3a_0}$$

PERTURBATION THEORY

$$E_n^{(1)} = H'_{nn} \qquad E_n^{(2)} = \sum_{m \neq n} \frac{|H'_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$\left|\psi_n^{(1)}\right\rangle = \sum_{m \neq n} \frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}} \left|\psi_m^{(0)}\right\rangle$$